

# Obscuration to Clarity: Bone Suppression for Enhanced Localization in Pneumothorax Segmentation of Chest Radiographs

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## INTRODUCTION

- Pneumothorax segmentation in Chest X-Rays (CXRs) is limited by structured anatomical occlusions (ribs, clavicles), which overlap lung boundaries and degrade pixel and boundary-level localization
- Bone suppression methods:
  - Improve soft-tissue visibility<sup>[1]</sup>
  - Enhance detection/classification performance<sup>[2]</sup>
- The role of bone suppression in dense prediction tasks remains unexplored:
  - No evaluation for pixel-wise segmentation
  - No analysis of boundary-aware localization
- Does removing structured anatomical occlusions improve feature learning for precise spatial localization and segmentation, or only global classification performance?**

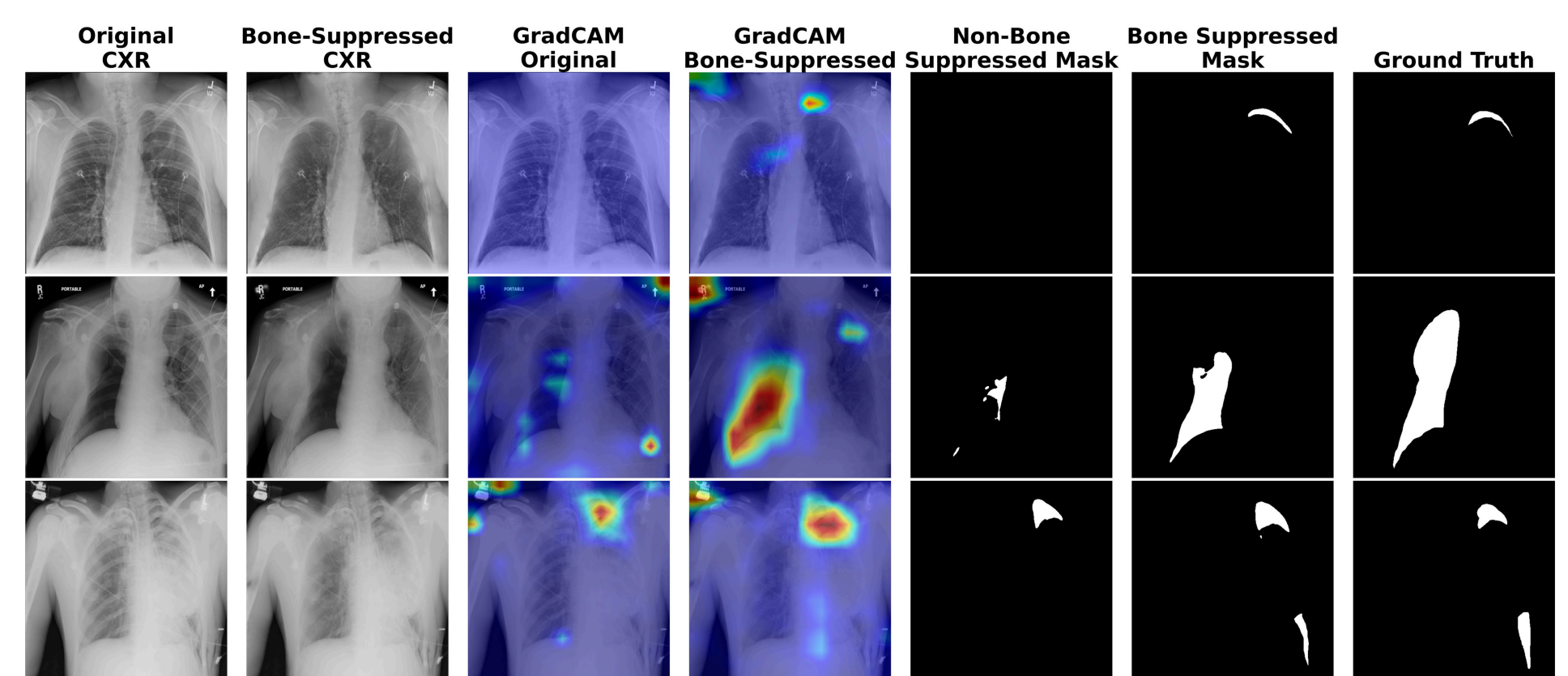
## OBJECTIVE

- To evaluate bone suppression as a **pre-processing step** for improving pixel-wise and boundary-level pneumothorax segmentation
- To systematically benchmark across:
  - 2 Datasets:** SIIM-ACR, PTX-498
  - 3 Metrics:** DSC (Global Overlap), MASD & NSD (Boundary-aware Localization)
  - 4 Architectures:** UNet++, MANet, SegFormer, TransUNet
- Assess **boundary-level localization improvements**, not just global overlap or detection
- Test **architecture-agnostic robustness** of bone-suppressed CXRs
- Analyze whether gains extend to both segmentation and classification tasks, **reflecting enhanced clinical decision signals**

## RESULTS

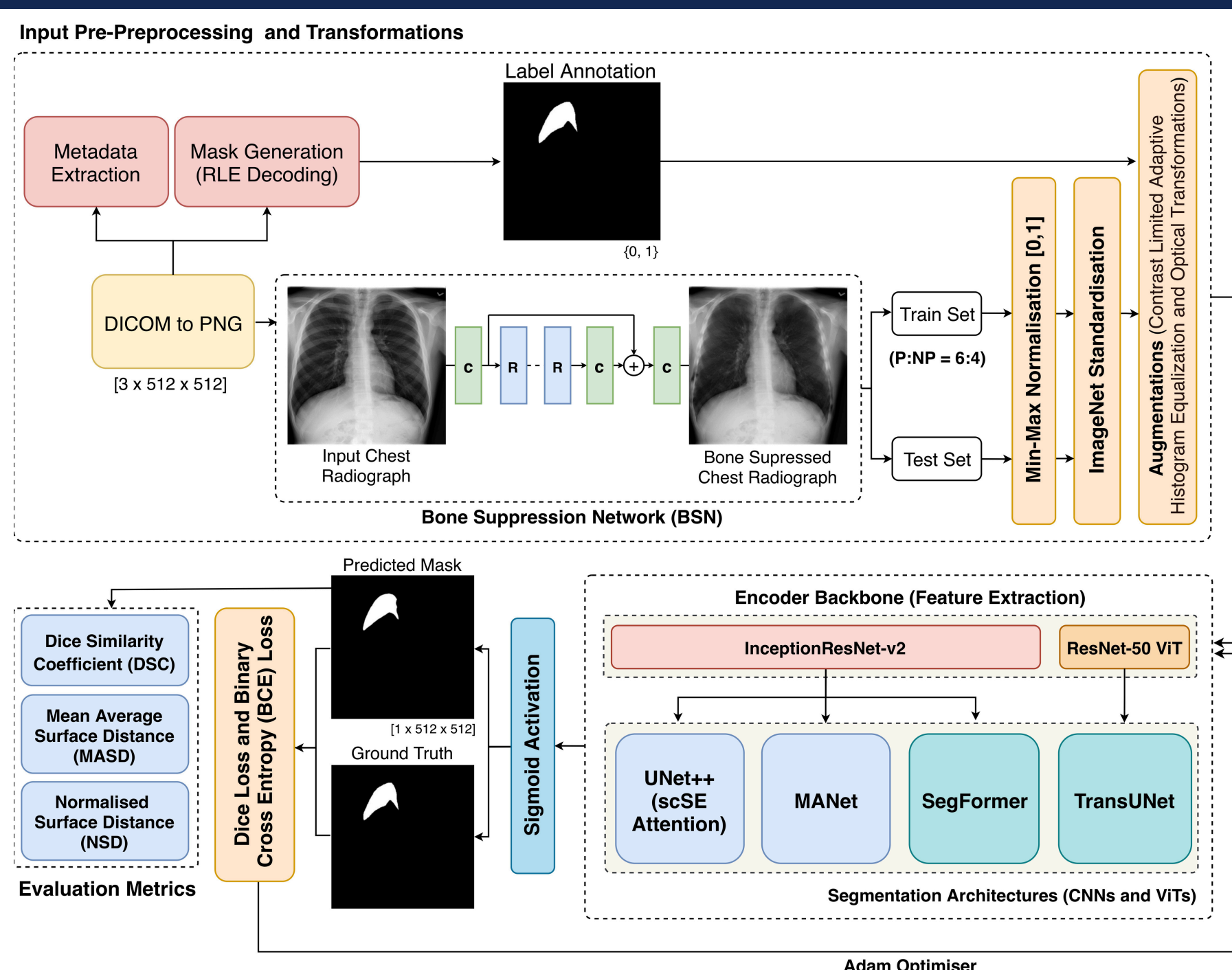
| Architecture | Encoder            | BS-CXR | DSC (↑)       | MASD (↓)      | NSD (↑)       | F1 (↑)        | MCC (↑)       |
|--------------|--------------------|--------|---------------|---------------|---------------|---------------|---------------|
| UNet++       | Inception-ResNetV2 | —      | 0.8064        | 5.4757        | 0.7673        | 0.7695        | 0.7004        |
|              |                    | ✓      | <b>0.8317</b> | <b>4.5423</b> | <b>0.7903</b> | <b>0.7933</b> | <b>0.7340</b> |
| MANet        | Inception-ResNetV2 | —      | 0.7830        | 5.4116        | 0.7397        | 0.7355        | 0.6569        |
|              |                    | ✓      | <b>0.8213</b> | <b>4.8838</b> | <b>0.7832</b> | <b>0.7788</b> | <b>0.7141</b> |
| SegFormer    | Inception-ResNetV2 | —      | 0.8088        | 5.3425        | 0.7665        | 0.7577        | 0.6866        |
|              |                    | ✓      | <b>0.8288</b> | <b>5.0721</b> | <b>0.7882</b> | <b>0.8065</b> | <b>0.7496</b> |
| TransUNet    | ResNet50-ViT       | —      | 0.7987        | 5.6073        | 0.7594        | 0.7496        | 0.6744        |
|              |                    | ✓      | <b>0.8342</b> | <b>5.0335</b> | <b>0.7890</b> | <b>0.7973</b> | <b>0.7382</b> |

Bone suppression yields consistent, statistically significant improvements ( $p < 0.05$ ) across all architectures, with **↑ DSC (+3.7%)**, **↑ NSD (+3.9%)**, and **↓ MASD (-10.6%)**, including up to **17% reduction in boundary error**, establishing BS-CXR as a universal input-level enhancement for precise pneumothorax segmentation.



**Grad-CAM** visualizations show sharper, pathology-focused activations with **reduced rib-induced noise**, leading to cleaner boundary delineation, **fewer false positives**, and **improved downstream classification (+9.5% MCC, +6.4% F1)**, reinforcing clinical reliability of segmentation-driven diagnosis.

## METHODOLOGY



- Two-Stage Cascaded Pipeline:** ResNet-based BSN pre-processing suppresses structured bony occlusions, followed by segmentation on BS-CXR for enhanced soft-tissue localization
- Hybrid Boundary-Aware Optimization:** Models are trained using a joint Dice-BCE Loss  $L_{Dice-BCE} = \alpha_D \cdot L_{Dice} + \alpha_B \cdot L_{BCE}$  to jointly optimize region overlap and boundary delineation

## CONCLUSION

- We show that removing structured anatomical occlusions at the input level fundamentally improves feature representations for boundary-sensitive pneumothorax segmentation, addressing a key limitation of 2D CXR imaging
- Clinical Relevance:** Improved pixel-level localization translates to more reliable detection and triaging, supporting accurate severity assessment of pneumothorax in automated clinical workflows
- Future Work:** End-to-end joint optimization, scaling to **larger multi-center datasets**, and validation across **broader thoracic pathologies** to strengthen generalizability and clinical deployment

## REFERENCES

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